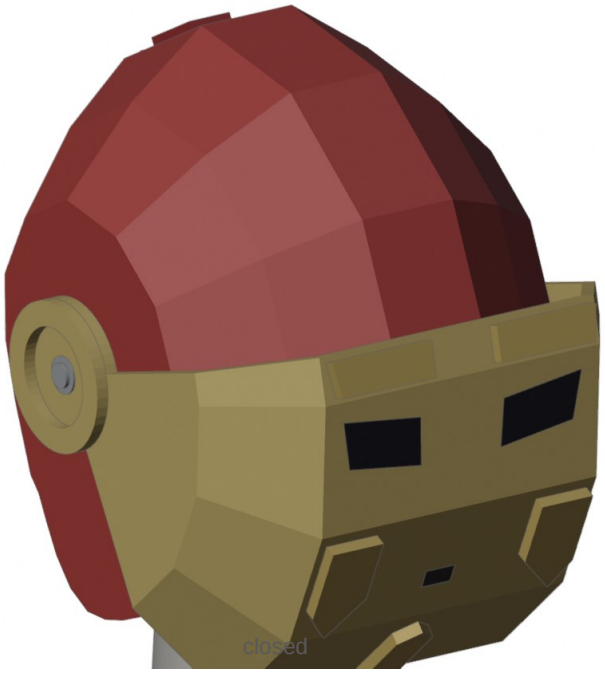
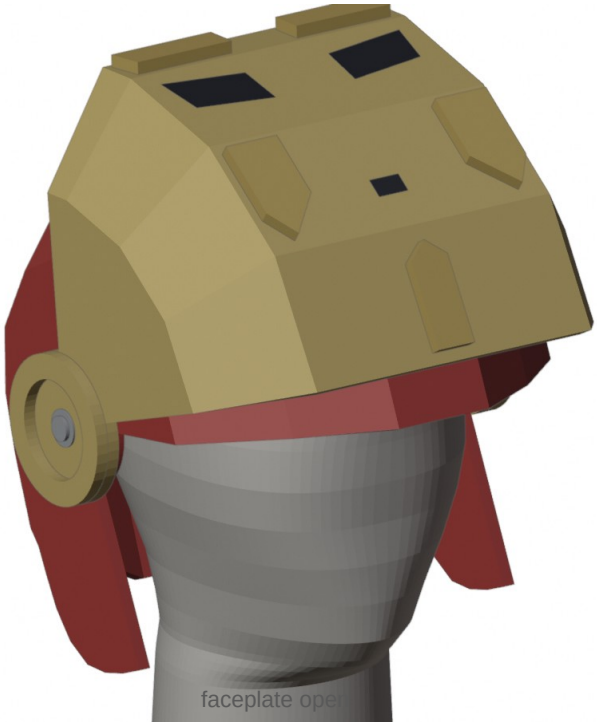
CARDBOARD HELMET

Flip-up faceplate on ear pivots | flat panels, numbered edges | full-size stencils from a 3D model

Fitted to: forehead 55.0 cm around, chin-over-top-to-nape 66.0 cm, 6 mm board. Outside: about 234 mm wide.





1. CHECK THE PRINT FIRST

Print every page on US Letter at 100% / Actual size (turn off 'fit to page'). Measure the bar at the bottom of any page: it must be exactly 100 mm. If it isn't, fix the printer setting before cutting anything.

2. WHAT YOU NEED

- 6 mm double-wall corrugated cardboard: about 0.24 m² of panels plus offcuts (large appliance or moving boxes)
- Glue stick or spray adhesive (to stick the 9 stencil pages on)
- Hot glue gun and plenty of sticks; masking tape
- Craft knife with fresh blades, steel ruler, cutting mat
- 2 Chicago screws (binding posts), flat heads: 5 mm diameter, 14 mm post (shorter than the 15 mm they clamp, so they can be tightened); removable thread locker
- 6 fender washers for 5 mm screws: about 18 mm across, 1 mm thick
- Optional: thin foam for padding (up to 8 mm keeps the fit)

3. CHECKED ON THE 3D MODEL BEFORE PRINTING

PASS Head fit: at least 12.0 mm between your head and the inside of the helmet (target 12); 17 mm above the crown

PASS Ears: 8.0 mm from the side plates, 5.0 mm from the screw heads inside them

PASS Face: faceplate at least 17.0 mm from your face (target 15; nose 26, chin 21 mm)

PASS Faceplate swing: tested at 161 angles from closed to 80 deg: no contact; tightest gap 5.5 mm (arms ride 1.0 mm off the side plates on the washers); never closer than 17 mm to your head

PASS Vision: both eyes look straight out through the slots; each eye sees -5 to 25 deg sideways and -5 to 10 deg up/down

PASS No gaps: 1728 sight lines from inside all hit the helmet

PASS Putting it on: a collision-free path exists to lower it onto your head (faceplate open)

PASS Flat panels: every panel flat to 3e-14 mm; normals consistent

PASS Pieces match the model: every flattened panel within 9e-14 mm of its 3D shape; 136 seams each cut on both sides with matching lengths

PASS Joints: every seam turns outwards (up to 56 deg), so square-cut edges meet at the outer corner and leave a V of at most 5.7 mm inside for hot glue (limit 8)

PASS Slots: eye and mouth slots leave at least 7.0 mm of board to every panel edge (limit 6)

PASS Pages: 9 stencil pages; every piece inside the printable area, none touching

PASS Printed PDF re-measured: 2350 cut lines read back from this PDF match the model within 0.001 mm; 100 mm bars measure 100.00 mm

this bar must measure exactly 100 mm

cut straight down, blade upright

12 edges with the same number are glued together, outer corners touching

Print at 100% / Actual size. Printed side = INSIDE of the helmet.

CARDBOARD HELMET - HOW TO BUILD

Read once before cutting. Piece names are printed on every stencil.

1 STICK, THEN CUT

Glue each printout onto the board with the printed side up. The printed side becomes the INSIDE of the helmet, so the outside stays clean for paint. Cut every solid line in several light passes with the blade held upright, so the edges come out square. Cut the eye and mouth slots and the pivot holes too.

2 HOW THE JOINTS WORK

Every edge has a number, and the edge with the same number on another panel is its partner. Hold the two panels printed side down with those edges touching and tape along the seam on the outside. Fold them away from the tape: the outer corners stay together and a V opens on the inside. Fill it with hot glue.

3 SHELL

Join the centre panels S-C-1 to S-C-8 front to back. Add the S-R1 and S-L1 panels alongside, then S-R2 and S-L2, then the side plates S-SIDE-R and S-SIDE-L. Each panel sets its neighbours' angles, so the shape locks in as the rows close up.

4 TRY THE SHELL ON

Before going further, put the shell on. There should be about a finger's width all round. The front edge sits just above your brows.

5 FACEPLATE

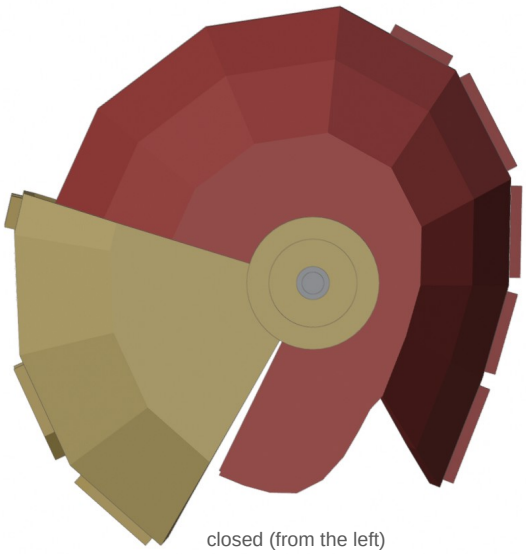
Join F-C-1 to F-C-4, then the F-R1 and F-L1 panels (with the eye slots), then F-R2 and F-L2, then F-R3 and F-L3, then the pivot arms F-ARM-R and F-ARM-L.

6 PIVOTS

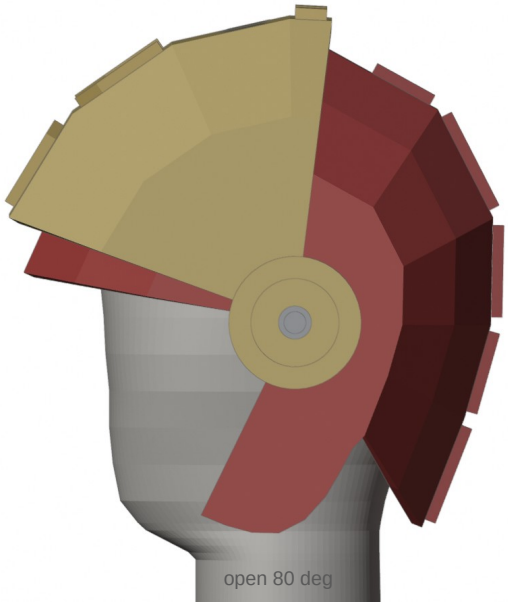
Put a washer on a Chicago screw post and push it from the inside through the side plate. Add a washer, the arm and a washer, then screw on the cap with a drop of thread locker. Tighten until it bottoms out: that is as tight as it goes, and it stops before it crushes the board.

7 OVERLAYS AND FINISH

Glue an E-RING round each screw cap, then the overlays (positions on the next page). Tape over the cut edges so the flutes don't show, prime, then paint. Swing the faceplate fully open and closed to check nothing rubs.

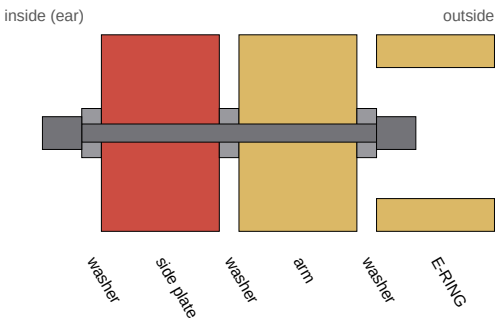


closed (from the left)



PIVOT STACK (section through the pivot)

Chicago screw: 14 mm post, clamps 15 mm



this bar must measure exactly 100 mm

cut straight down, blade upright

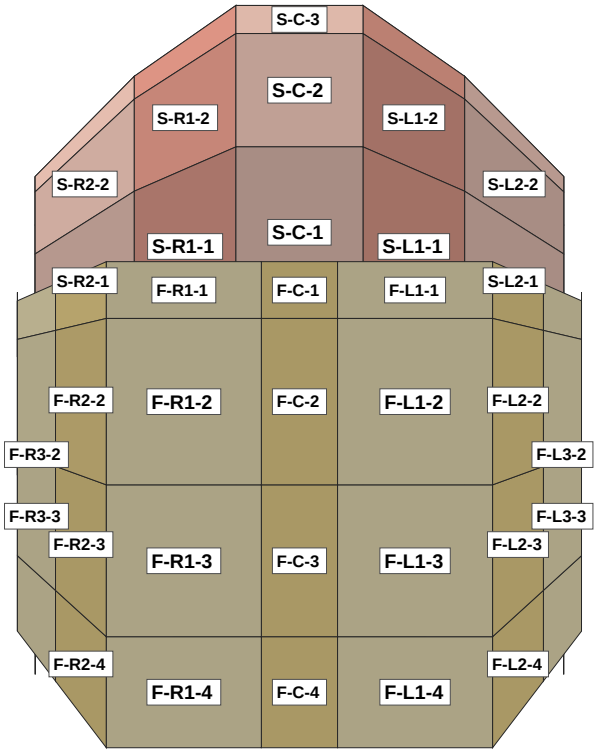
12 edges with the same number are glued together, outer corners touching

Print at 100% / Actual size. Printed side = INSIDE of the helmet.

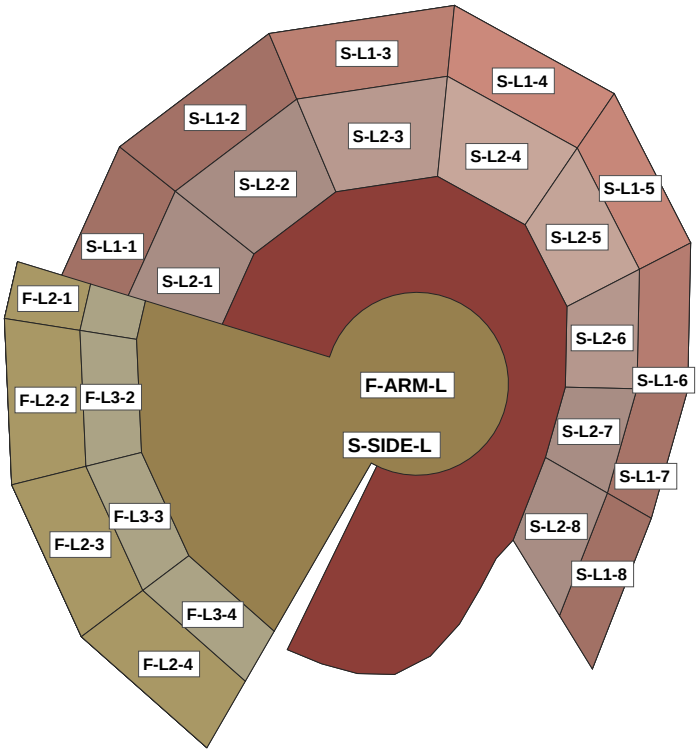
CARDBOARD HELMET - WHERE EACH PIECE GOES

Outside views of the finished helmet. Seam numbers on the stencils say exactly which edges meet.

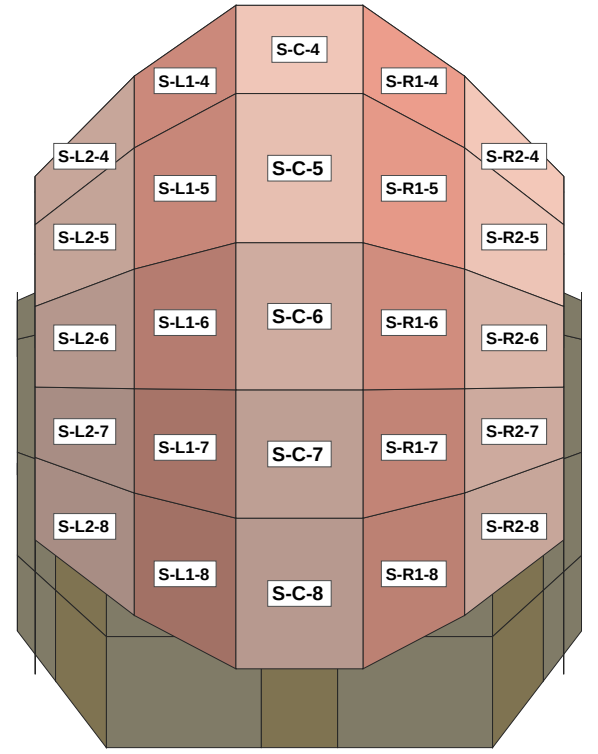
FRONT



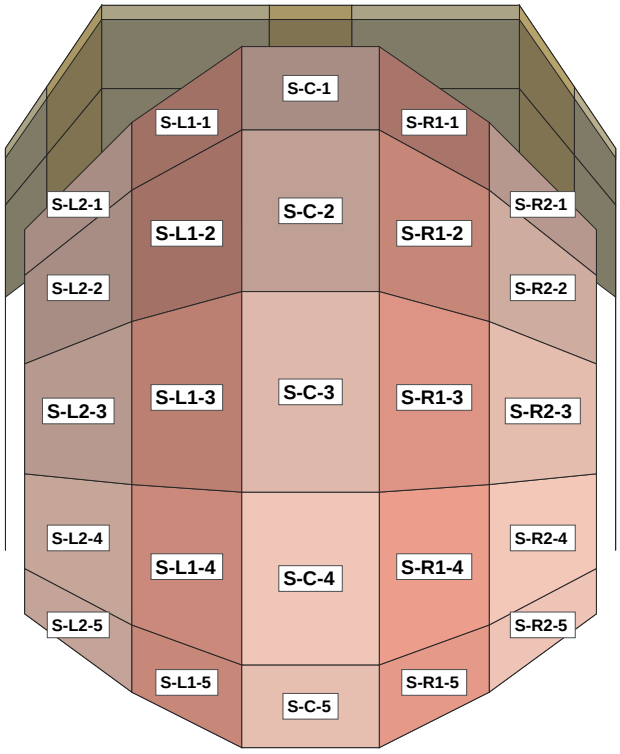
LEFT SIDE



BACK



TOP (front is down)



Overlays (centred on their panel): O-BROW-L on F-L1-1, O-BROW-R on F-R1-1, O-CHEEK-L on F-L1-3, O-CHEEK-R on F-R1-3, O-CHIN on F-C-4, O-CREST-1 on S-C-4, O-CREST-2 on S-C-5, O-CREST-3 on S-C-6, O-CREST-4 on S-C-7, O-CREST-5 on S-C-8.

this bar must measure exactly 100 mm

cut straight down, blade upright

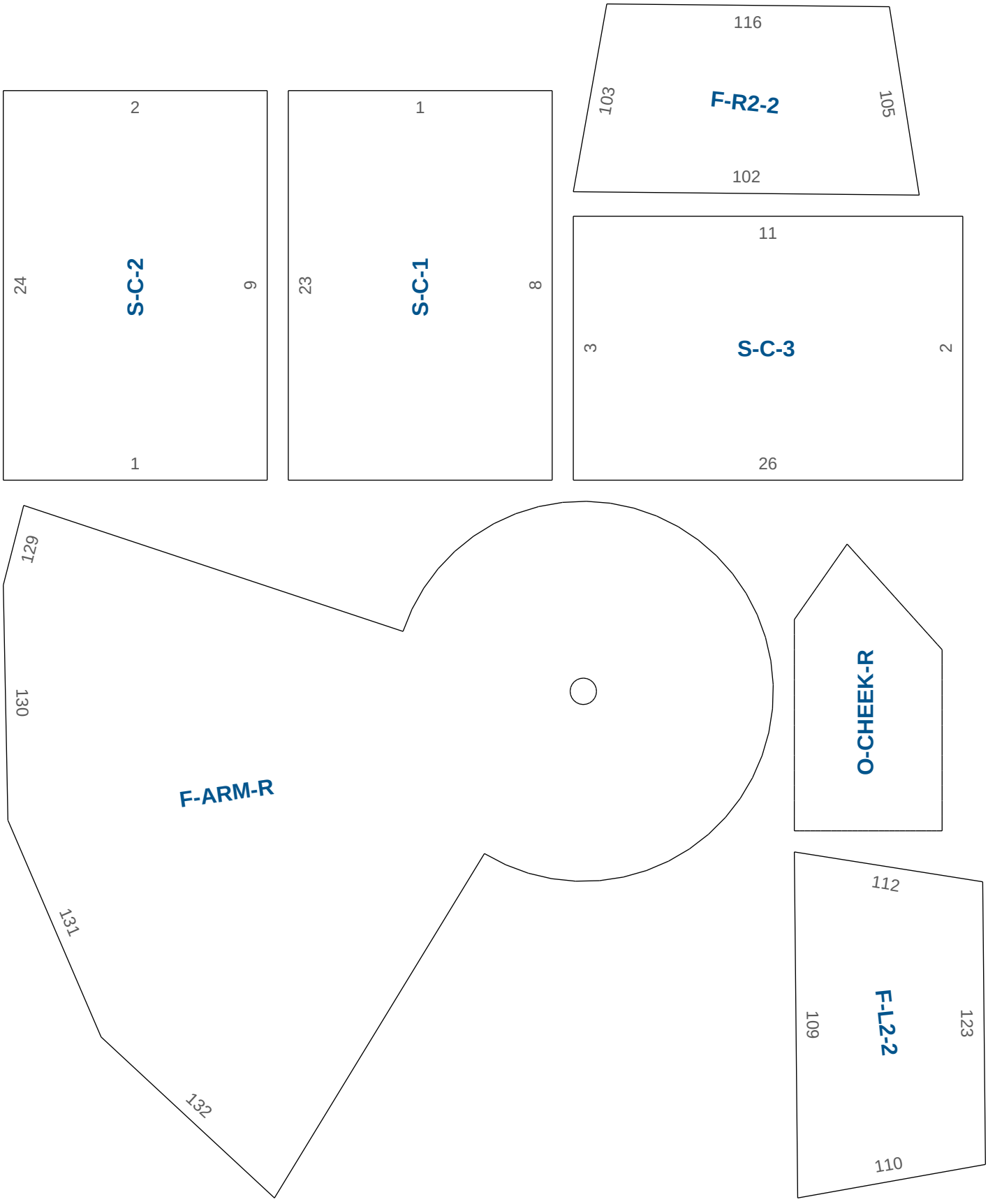
12 edges with the same number are glued together, outer corners touching

Print at 100% / Actual size. Printed side = INSIDE of the helmet.

GUIDE - DO NOT CUT

CARDBOARD HELMET - STENCILS

Pieces: S-C-1, S-C-2, S-C-3, F-R2-2, F-L2-2, F-ARM-R, O-CHEEK-R



this bar must measure exactly 100 mm

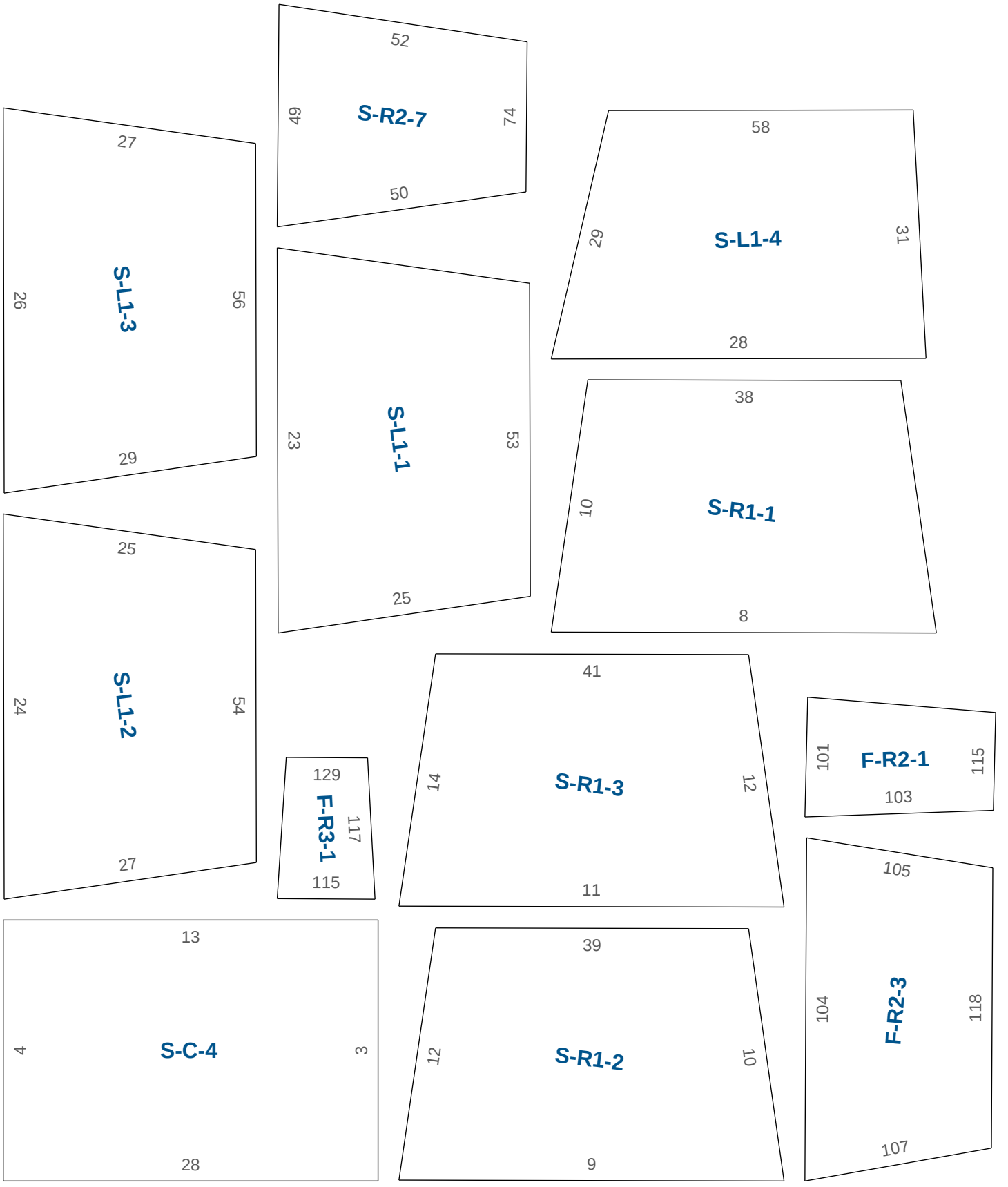
cut straight down, blade upright

12 edges with the same number are glued together, outer corners touching

Print at 100% / Actual size. Printed side = INSIDE of the helmet.

CARDBOARD HELMET - STENCILS

Pieces: S-C-4, S-R1-1, S-R1-2, S-R1-3, S-L1-1, S-L1-2, S-L1-3, S-L1-4, S-R2-7, F-R2-1, F-R2-3, F-R3-1



this bar must measure exactly 100 mm

cut straight down, blade upright

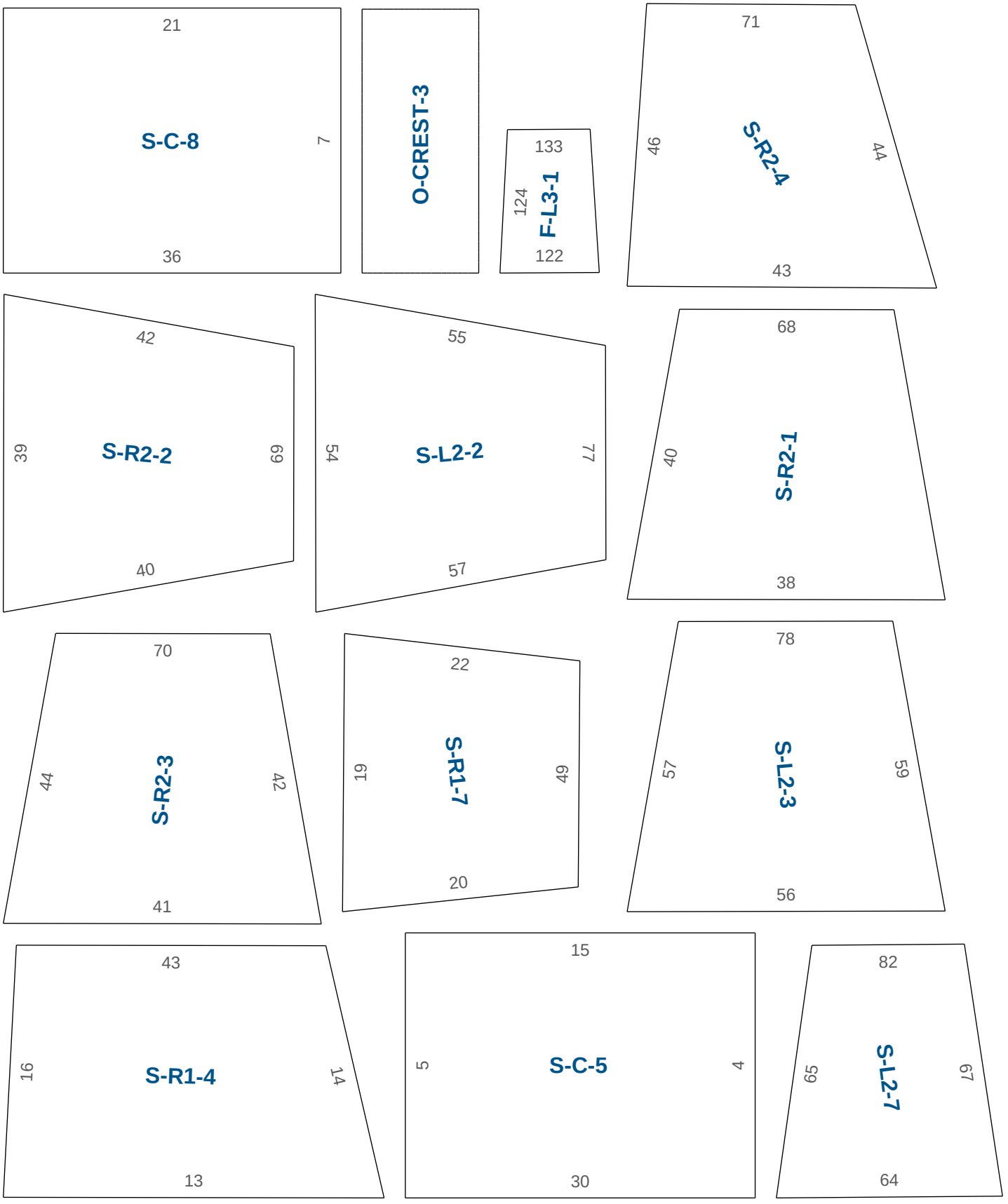
12 edges with the same number are glued together, outer corners touching

Print at 100% / Actual size. Printed side = INSIDE of the helmet.

CUT PATTERN - ACTUAL SIZE

CARDBOARD HELMET - STENCILS

Pieces: S-C-5, S-C-8, S-R1-4, S-R1-7, S-R2-1, S-R2-2, S-R2-3, S-R2-4, S-L2-2, S-L2-3, S-L2-7, F-L3-1, O-CREST-3



this bar must measure exactly 100 mm

cut straight down, blade upright

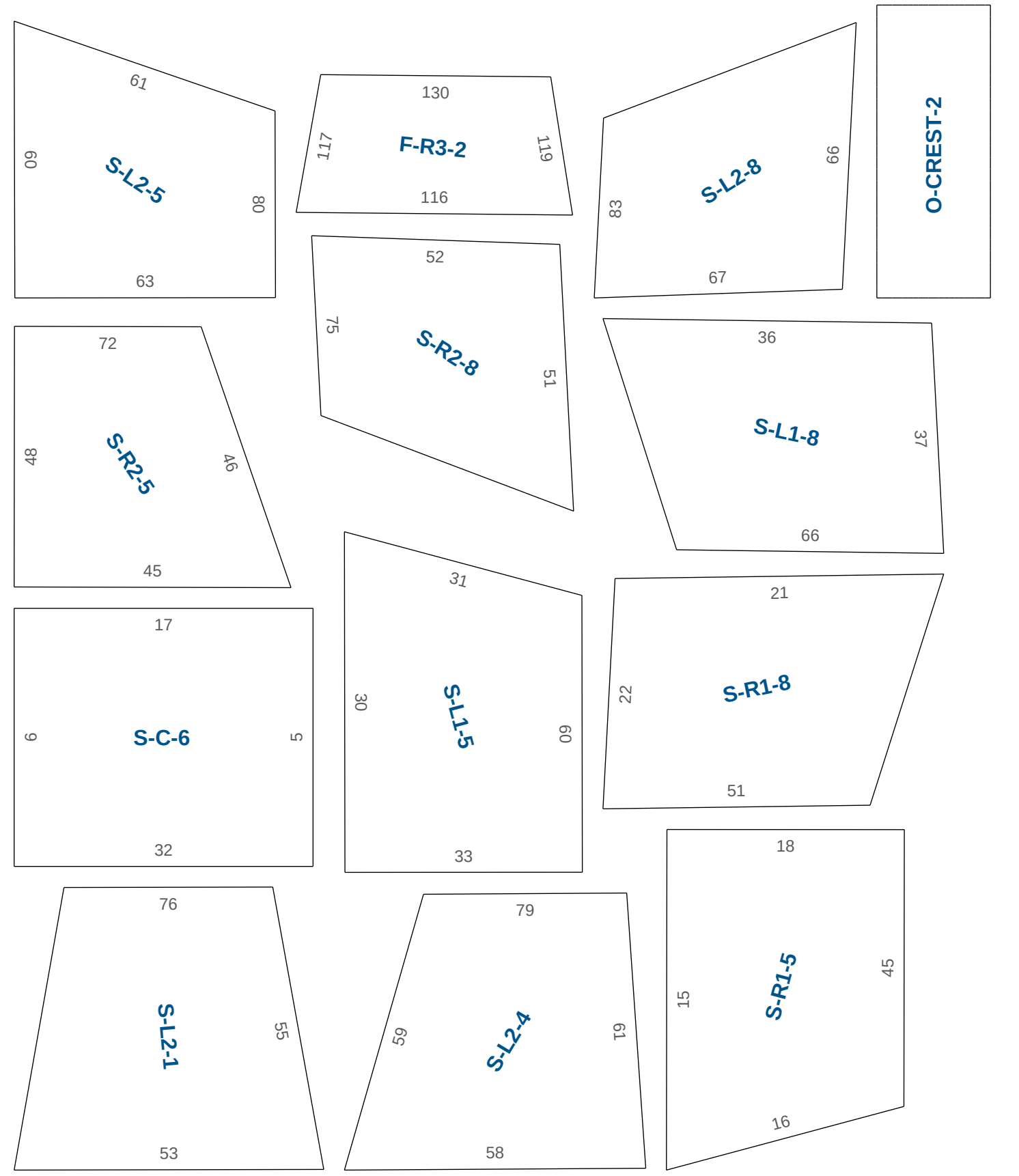
12 edges with the same number are glued together, outer corners touching

Print at 100% / Actual size. Printed side = INSIDE of the helmet.

CUT PATTERN - ACTUAL SIZE

CARDBOARD HELMET - STENCILS

Pieces: S-C-6, S-R1-5, S-R1-8, S-L1-5, S-L1-8, S-R2-5, S-R2-8, S-L2-1, S-L2-4, S-L2-5, S-L2-8, F-R3-2, O-CREST-2



this bar must measure exactly 100 mm

cut straight down, blade upright

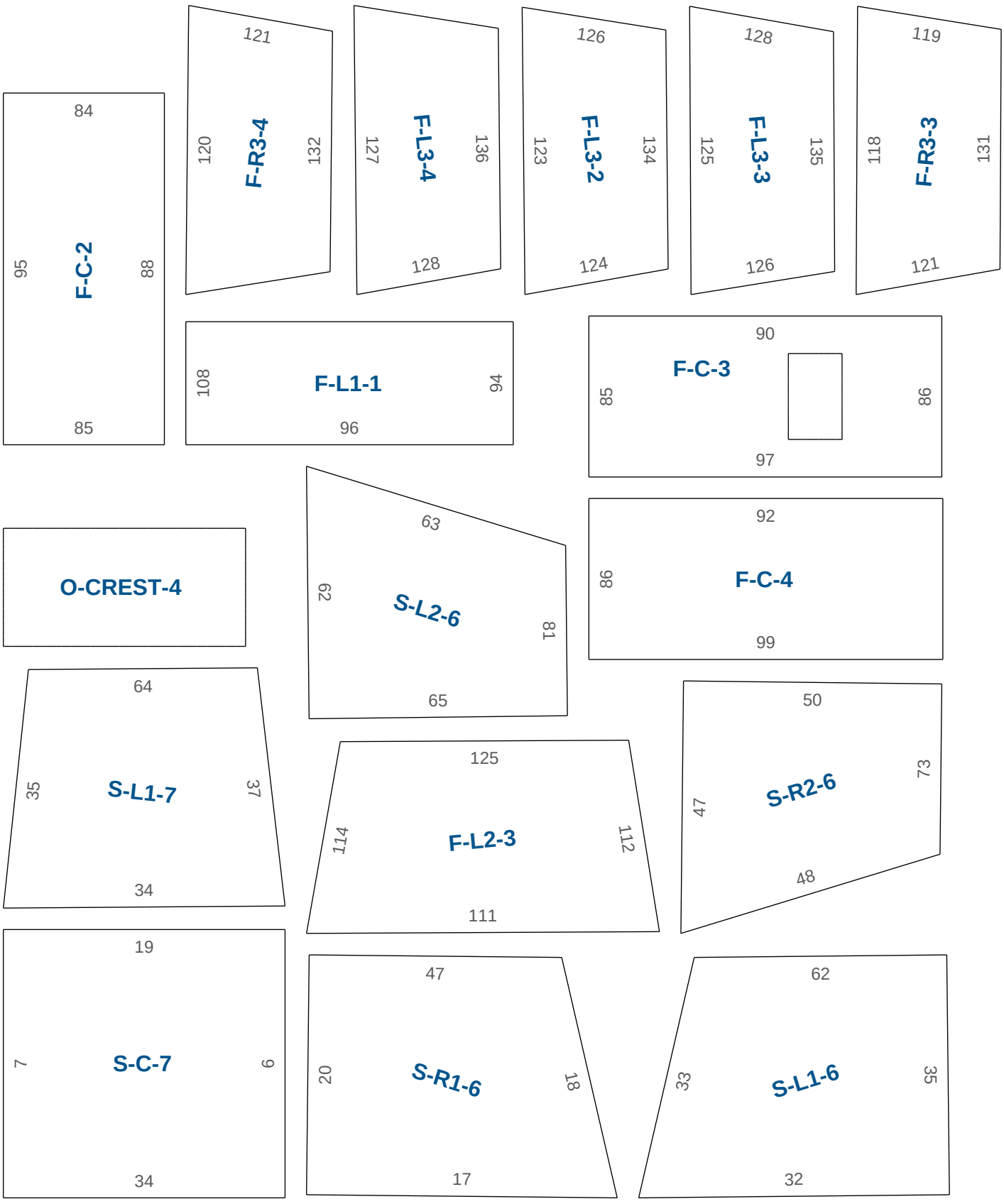
12 edges with the same number are glued together, outer corners touching

Print at 100% / Actual size. Printed side = INSIDE of the helmet.

CUT PATTERN - ACTUAL SIZE

CARDBOARD HELMET - STENCILS

Pieces: S-C-7, S-R1-6, S-L1-6, S-L1-7, S-R2-6, S-L2-6, F-C-2, F-C-3, F-C-4, F-L1-1, F-L2-3, F-R3-3, F-R3-4, F-L3-2, F-L3-3, F-L3-4, O-CREST-4



this bar must measure exactly 100 mm

cut straight down, blade upright

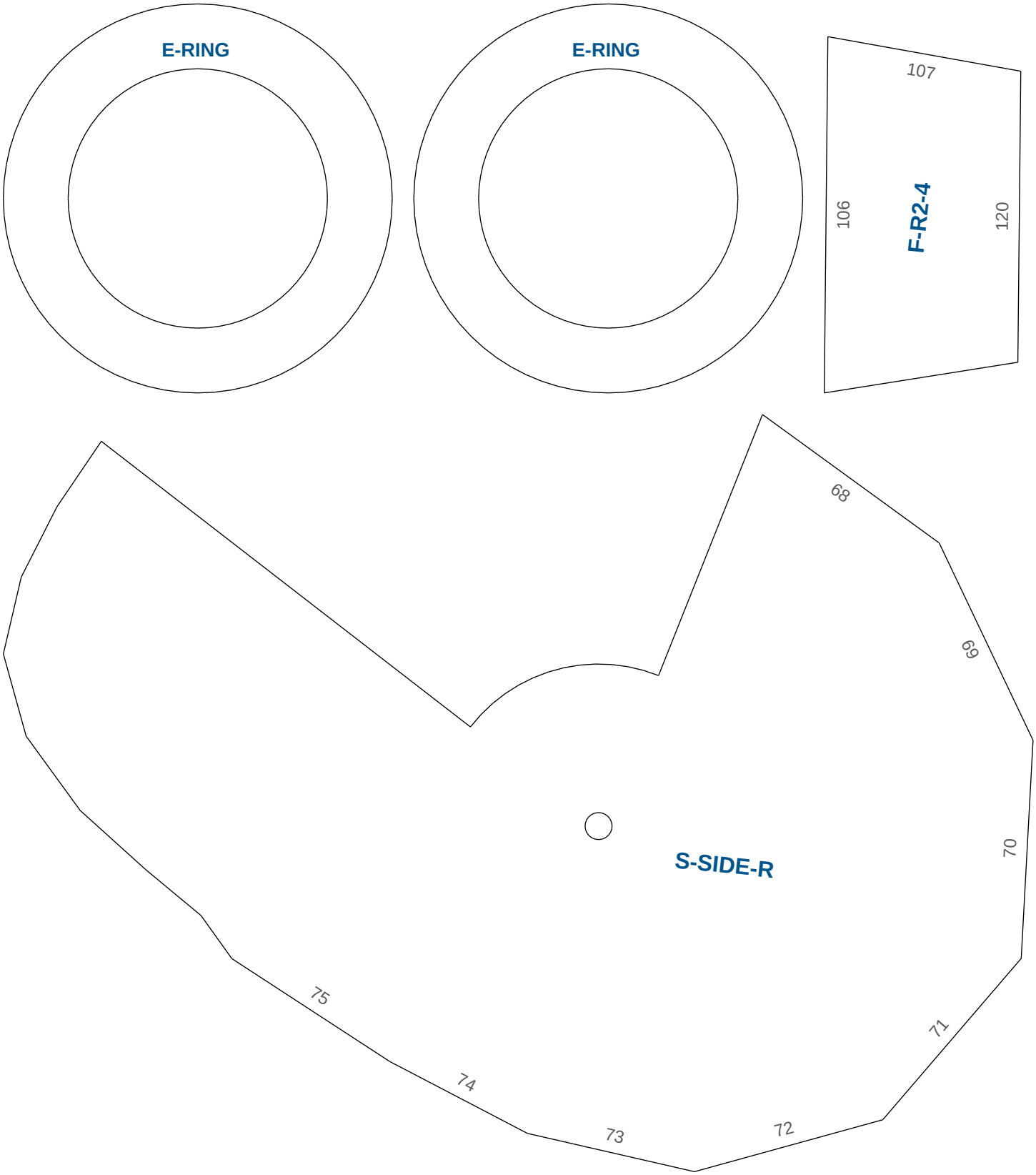
12 edges with the same number are glued together, outer corners touching

Print at 100% / Actual size. Printed side = INSIDE of the helmet.

CUT PATTERN - ACTUAL SIZE

CARDBOARD HELMET - STENCILS

Pieces: S-SIDE-R, F-R2-4, E-RING (1/2), E-RING (2/2)



this bar must measure exactly 100 mm

cut straight down, blade upright

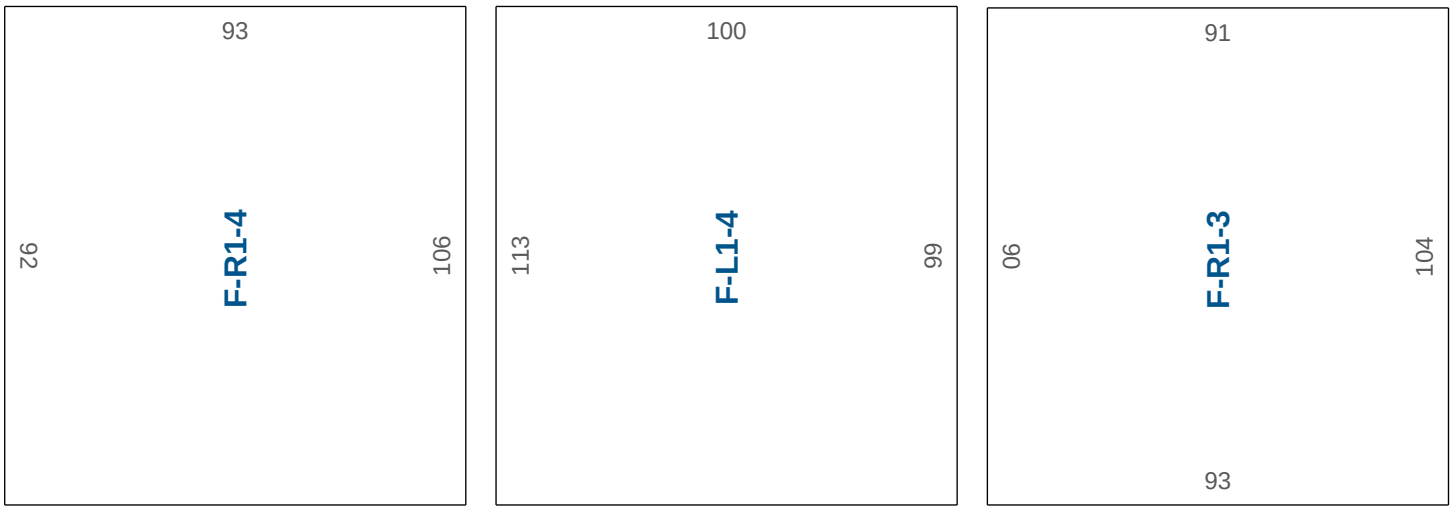
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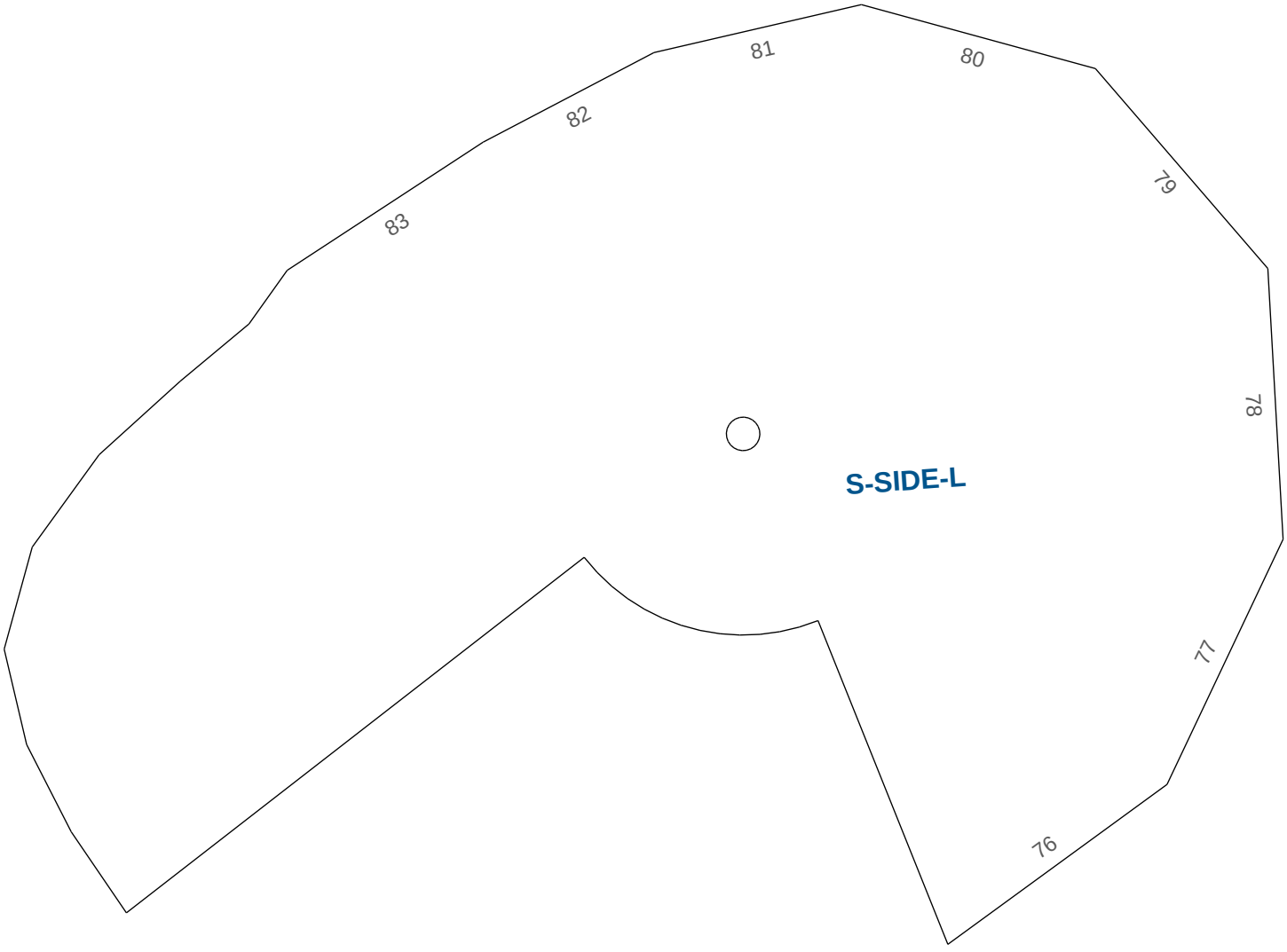
Print at 100% / Actual size. Printed side = INSIDE of the helmet.

CUT PATTERN - ACTUAL SIZE

CARDBOARD HELMET - STENCILS

Pieces: S-SIDE-L, F-R1-3, F-R1-4, F-L1-4





this bar must measure exactly 100 mm

cut straight down, blade upright

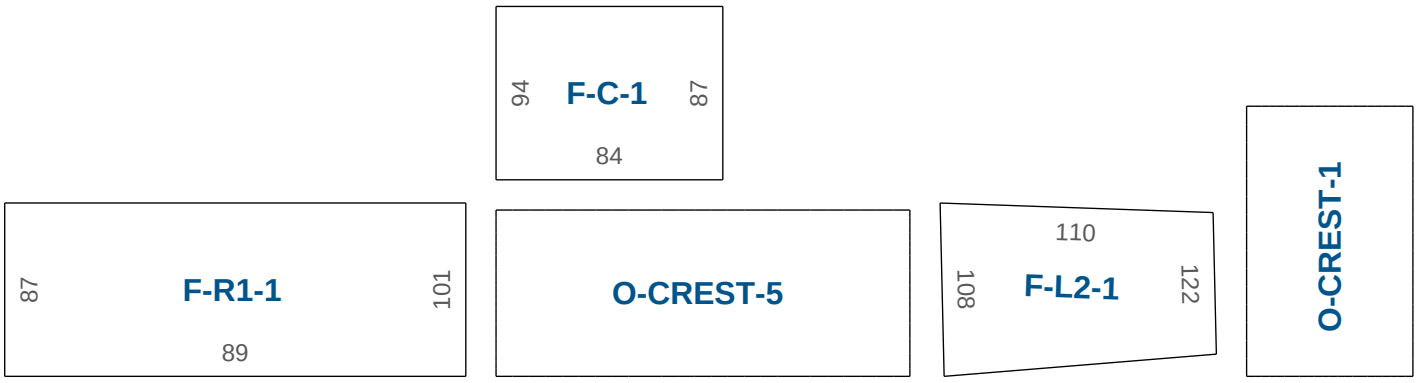
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Print at 100% / Actual size. Printed side = INSIDE of the helmet.

CUT PATTERN - ACTUAL SIZE

CARDBOARD HELMET - STENCILS

Pieces: F-C-1, F-R1-1, F-L2-1, O-CREST-1, O-CREST-5



this bar must measure exactly 100 mm

cut straight down, blade upright

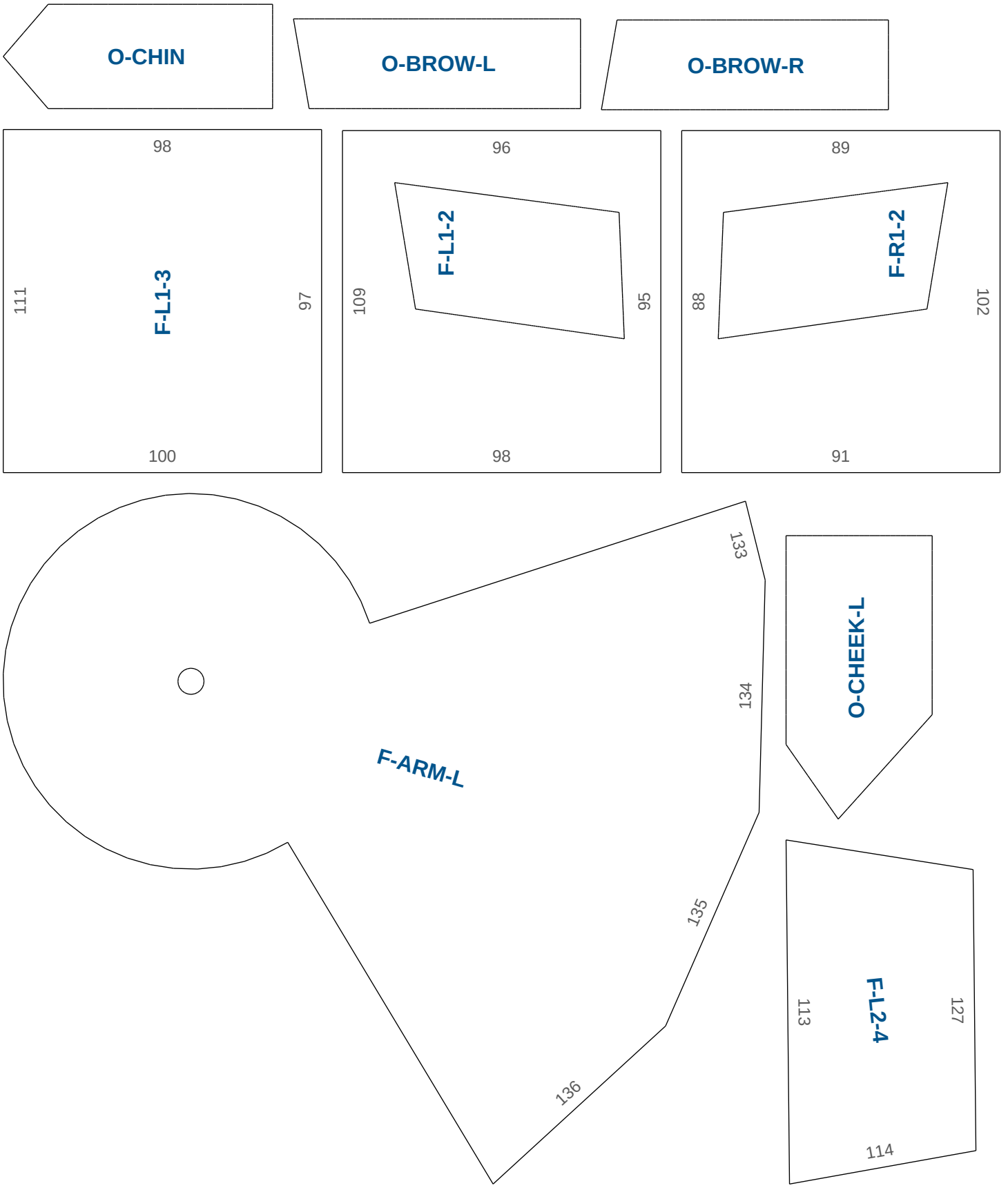
12 edges with the same number are glued together, outer corners touching

Print at 100% / Actual size. Printed side = INSIDE of the helmet.

CUT PATTERN - ACTUAL SIZE

CARDBOARD HELMET - STENCILS

Pieces: F-R1-2, F-L1-2, F-L1-3, F-L2-4, F-ARM-L, O-BROW-L, O-BROW-R, O-CHEEK-L, O-CHIN



this bar must measure exactly 100 mm

cut straight down, blade upright

12 edges with the same number are glued together, outer corners touching

Print at 100% / Actual size. Printed side = INSIDE of the helmet.

CUT PATTERN - ACTUAL SIZE